Continuous integer cocycles: the broad evaluation formula fails

Result. Lean disproves the evaluation-surjectivity required by the broad arbitrary-discrete-module wording of Bruin2011 equation2.1. The example uses the actual Krull absolute Galois group of F2, actual square Frobenius, and discrete integers with trivial continuous action. Every continuous one-cocycle is zero; the actual coinvariant quotient contains the nonzero integer class1, which evaluation cannot reach. The finite-module Lemma2.1/application is retained. [pcc:unverified]

Actual source group	Absolute Gal(F2bar/F2), with compact Krull topology Exact q-power Frobenius: $x \mapsto x^2$
Actual coefficient group	Discrete integers, trivial continuous action Every continuous 1-cocycle vanishes
Actual evaluation	Coinvariants are integers; integer class 1 is nonzero Evaluation misses class 1, even on any class quotient

Proof and faithful domains. A continuous homomorphism from a compact group has finite image in a discrete group. Image elements have finite order, so a torsion-free target forces all values to be identity. Lean constructs the actual Frobenius AlgEquiv and proves its x^2 formula, verifies trivial additive-action identity/composition laws and joint continuity, and proves every continuous integer-valued function satisfying the standard cocycle identity vanishes. Bundling as continuous homomorphisms excludes no other cocycles. [pcc:unverified]

The claimed map cannot be repaired by quotienting. The actual image of identity minus identity consists only of zero. Lean proves the integer class1 remains nonidentity in the actual coinvariant quotient and that evaluation is nonsurjective at every actual Galois element, specifically at square Frobenius. It directly rules out any type H with a surjective cocycle-representative map and a bijection H→coinvariants agreeing with evaluation. No homemade H1 object or supplied cohomology value is used. [pcc:unverified]

Controls and notation. Integer1 is not killed by any positive natural number. The identity continuous homomorphism on discrete integers has nonidentity value at1: continuity alone does not force vanishing after removing compactness. Code uses Multiplicative$\mathbb{Z}$ for the quotient API; its multiplicative identity denotes additive integer0. The missing class represents additive integer1. [pcc:unverified]

Primary review and limit. Bruin2011, JTNB23,323–328, published325/PDF4, equation2.1: the preceding module/action paragraph quantifies arbitrary discrete groups with continuous action. The finite restriction enters the subsequent Lemma2.1. Prior full published-note reading/allseven-page inspection credited. Selected author preprint page3/equation1 context read/rendered/fullyinspected, with matching wording. Bounded correction/qualification search and author links checked; no novelty/global errata-absence claim. No cited Serre theorem is marked read or refuted. [pcc:unverified]

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